

References

- [1] Panagiotopoulou, A., Anastassopoulos, V.(2008).Super-resolution reconstruction of thermal infrared images.
- [2] Gade, R., & Moeslund, T. B. (2014). Thermal cameras and applications: a survey. *Machine vision and applications*, 25(1), 245-262.
- [3] Deepak, S., Sahoo, S., & Patra, D. (2021, December). Super-resolution of thermal images using GAN network. In *2021 Advanced Communication Technologies and Signal Processing (ACTS)* (pp. 1-5). IEEE.
- [4] Rivadeneira, R. E., Suárez, P. L.,Sappa, A. D., & Vintimilla, B. X. (2019,August). Thermal image superresolution through deep convolutional neural network. In *International conference on image analysis and recognition* (pp. 417-426). Cham: Springer International Publishing.
- [5] Rivadeneira, R. E., Sappa, A. D., & Vintimilla, B. X. (2020, February). Thermal image super-resolution: A novel architecture and dataset. In *VISIGRAPP (4: VISAPP)* (pp. 111-119).
- [6] Suárez, P. L., Carpio, D., & Sappa, A. D. (2024). Enhancement of guided thermal image super-resolution approaches. *Neurocomputing*, 573, 127197.
- [7] Jiang, H., & Chen, Z. (2024). Flexible window-based self-attention transformer in thermal image super-resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 3076-3085).
- [8] Dong, Y. C., Jiang, J., Wang, Q. L., Chen, W., & Ye, J. H. (2024). SUPER-RESOLUTION RECONSTRUCT ION AND HIGH-PRECISION TEMPERATURE MEASUREMENT OF THERMALIMAGES UNDERHIGH-TEMPERATURE SCENES BASED ON NEURAL NETWORK. *Advanced Steel Construction*, 20(2), 169-178.
- [9] Kansal, P., & Nathan, S. (2025). Dual-Input Frequency-Aware Network for High-Quality Thermal Image Super-Resolution. In *Proceedings of the Computer Vision and Pattern RecognitionConference* (pp.4610-4620).
- [10] Huang, Y., Miyazaki, T., Liu, X., & Omachi, S. (2025). Infrared Image Super-Resolution: A Systematic Review and Future Trends. *IEEE journal of selected topics in applied earth observations and remote sensing*.
- [11] Zhong, H., Wang, Y., & Zhao, S. (2025). SwinPaste: A Swin Transformer-Based Framework for RGB-Guided Thermal Image Super-Resolution. In *Proceedings of the Computer Vision and Pattern Recognition Conference* (pp. 4589-4594).
- [12] Rafael E. Rivadeneira, Angel D. Sappa, Riad Hammoud, Jiyong Rao, Hang Zhong, Yu Wang, Shengjie Zhao, Zhiwei Zhong, Yung-Hui Li, Shiqi Wang, Qiangqiang Shen, Hanzhang Wang, Xuanqi Zhang; Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, 2025, pp. 4660-4669
- [13] Zhang, X., Li, C., Meng, Q., Liu, S., Zhang, Y., & Wang, J. (2018). Infrared image super resolution by combining compressive sensing and deep learning. *Sensors*, 18(8), 2587.]
- [14] Zoetgnande, Y. W. K., Dillenseger, J. L., & Alirezaie, J. (2019, July). Edge focused super-resolution of thermal images. In *2019 International Joint Conference on Neural Networks (IJCNN)* (pp. 1-8). IEEE.
- [15] Kwasniewska, A., Ruminski, J., Szankin, M., & Kaczmarek, M. (2020). Super-resolved thermal imagery for high-accuracy facial areas detection and analysis. *Engineering Applications of Artificial Intelligence*, 87, 103263.
- [16] Raimundo, J., Lopez-Cuervo Medina, S., Prieto, J. F., & Aguirre de Mata, J. (2021). Super resolution infrared thermal imaging using pansharpening

- algorithms: Quantitative assessment and application to UAV thermal imaging. *Sensors*, 21(4), 1265.
- [17] Kasliwal, A., Seth, P., Rallabandi, S., & Singhal, S. (2023). Corefusion: Contrastive regularized fusion for guided thermal super-resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 507-514).
- [18] Kasliwal, A., Gakhar, I., Kamani, A., Seth, P., & Verma, U. (2024). LapGSR: Laplacian Reconstructive Network for Guided Thermal Super-Resolution. *arXiv preprint arXiv:2411.07750*.
- [19] Arnold, C., Jouvét, P., & Seoud, L. (2024). SwinFuSR: an image fusion-inspired model for RGB-guided thermal image super-resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 3027-3036).
- [20] Salvetti, F., Mazzia, V., Khaliq, A., & Chiaberge, M. (2020). Multi-image super resolution of remotely sensed images using residual attention deep neural networks. *Remote Sensing*, 12(14), 2207.
- [21] Cascarano, P., Corsini, F., Gandolfi, S., Piccolomini, E. L., Mandanici, E., Tavasci, L., & Zama, F. (2020). Super-resolution of thermal images using an automatic total variation based method. *Remote Sensing*, 12(10), 1642.
- [22] Batchuluun, G., Lee, Y. W., Nguyen, D. T., Pham, T. D., & Park, K. R. (2020). Thermal image reconstruction using deep learning. *IEEE Access*, 8, 126839-126858.
- [23] Ren, Z., Zhao, J., Wang, C., Ma, X., Lou, Y., & Wang, P. (2022). Research on key technologies of super-resolution reconstruction of medium and long wave maritime infrared image. *Applied Sciences*, 12(21), 10871.
- [24] Chauhan, K., Patel, S. N., Kumhar, M., Bhatia, J., Tanwar, S., Davidson, I. E., ... & Sharma, R. (2023). Deep learning-based single-image super-resolution: A comprehensive review. *IEEE Access*, 11, 21811-21830.
- [25] Jiang, Y., Liu, Y., Zhan, W., & Zhu, D. (2023). Improved thermal infrared image super-resolution reconstruction method base on multimodal sensor fusion. *Entropy*, 25(6), 914.
- [26] Suárez, P. L., & Sappa, A. D. (2024). A Generative Model for Guided Thermal Image Super-Resolution. *Proceedings Copyright*, 765, 771.
- [27] Zou, Y., Chen, Z., Zhang, Z., Li, X., Ma, L., Liu, J., ... & Zhang, Y. (2026). Contourlet refinement gate framework for thermal spectrum distribution regularized infrared image super-resolution. *International Journal of Computer Vision*, 134(1), 23.
- [28] Chen, J., Tian, A., Chen, D., Guo, M., He, D., & Liu, Y. (2024). Super-resolution reconstruction method of the optical synthetic aperture image using generative adversarial network. *Open Physics*, 22(1), 20230194.
- [29] Didenko, A., Oliinyk, A., & Subbotin, S. (2023). Thermal Image Super-Resolution Methods Using Neural Networks. In *CMIS* (pp. 214-223).
- [30] Yun, H., Lo, S., Diepenbrock, C. H., Bailey, B. N., & Earles, J. M. (2024). VisTA-SR: Improving the Accuracy and Resolution of Low-Cost Thermal Imaging Cameras for Agriculture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 5470-5479).
- [31] Almasri, F., & Debeir, O. (2018, December). Multimodal sensor fusion in single thermal image super-resolution. In *Asian conference on computer vision* (pp. 418-433). Cham: Springer International Publishing.
- [32] Chudasama, V., Patel, H., Prajapati, K., Upla, K. P., Ramachandra, R., Raja, K., & Busch, C. (2020). Therisurnet-a computationally efficient thermal image super-resolution network. In *Proceedings of the IEEE/CVF*

- Conference on Computer Vision and Pattern Recognition Workshops* (pp. 86-87).
- [33] Prajapati, K., Chudasama, V., Patel, H., Sarvaiya, A., Upla, K. P., Raja, K., ... & Busch, C. (2021). Channel split convolutional neural network (ChaSNet) for thermal image super-resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 4368-4377).
- [34] Wang, K., Sun, Q., Wang, Y., Wei, H., Lv, C., Tian, X., & Liu, X. (2022). Cippsrnet: A camera internal parameters perception network based contrastive learning for thermal image super-resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 342-349).
- [35] Zhang, H., Hu, Y., & Yan, M. (2023). Thermal image super-resolution based on lightweight dynamic attention network for infrared sensors. *Sensors*, 23(21), 8717.
- [36] Rivadeneira Campodónico, R. E., & Vintimilla, B. X. (2023). *Thermal image super-resolution using deep learning techniques* (Doctoral dissertation, ESPOL. FIEC).
- [37] Liu, S., Yan, K., Qin, F., Wang, C., Ge, R., & Zhang, K. *Infrared image super-resolution via lightweight information split network* (2024).
- [38] Xu, Z., Gao, J., Wang, X., & Kang, C. (2024). Inf-OSRGAN: Optimized Blind Super-Resolution GAN for Infrared Images. *Applied Sciences*, 14(17), 7620.
- [39] Weng, Z., Liu, X., Liu, C., Guo, X., Shi, Y., & Lin, L. (2025). Dronesr: Rethinking few-shot thermal image super-resolution from drone-based perspective. *IEEE Sensors Journal*.
- [40] Rai, M., Maity, T., & Yadav, R. K. (2017). Thermal imaging system and its real time applications: a survey. *Journal of Engineering Technology*, 6(2), 290-303.
- [41] Hui, Z., Wang, X., & Gao, X. (2018). Fast and accurate single image super-resolution via information distillation network. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 723-731).
- [42] Almasri, F., & Debeir, O. (2018, November). RGB Guided Thermal Super-Resolution Enhancement. In *Cloudtech* (pp. 1-5).
- [43] Alfarzaei, M. S., Niu, Q., Zhao, J., Eshaq, R. M. A., & Hu, E. (2020). Coal/gangue recognition using convolutional neural networks and thermal images. *Ieee Access*, 8, 76780-76789.
- [44] Yang, Y., Li, Q., Yang, C., Fu, Y., Feng, H., Xu, Z., & Chen, Y. (2020). Deep networks with detail enhancement for infrared image super-resolution. *IEEE Access*, 8, 158690-158701.
- [45] Qian, X., Zhang, M., & Zhang, F. (2020). Sparse gans for thermal infrared image generation from optical image. *IEEE Access*, 8, 180124-180132.
- [46] Panda, G., Kundu, S., Bhattacharya, S., & Routray, A. (2023). LATIS: Lambda Abstraction-based Thermal Image Super-resolution. *arXiv preprint arXiv:2311.12046*.
- [47] Huang, Y., Miyazaki, T., Liu, X., & Omachi, S. (2025). Irsrmamba: Infrared image super-resolution via mamba-based wavelet transform feature modulation model. *IEEE Transactions on Geoscience and Remote Sensing*.
- [48] Shi, Y., Niu, S., Wang, L., Ye, L., Zhang, Y., & Hong, H. (2025). SFMHANet: surface fitting constrained multidimensional hybrid attention network for aero-optics thermal radiation effect correction. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*.